

Introduction to Machine Learning

Lab 1: Polynomial Regression

Hongteng Xu

1 Motivation

- Try to implement various training methods for polynomial regression.
- Implement commonly-used data normalization methods and check their influences on the mean-square-error (MSE) of testing data.

2 Tasks

1. Given the problem

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2, \quad (1)$$

where $\mathbf{X} = \{x_n^{d-1}\}_{n,d=1}^{N,D} \in \mathbb{R}^{N \times D}$ is the Vandermonde matrix, implement the function “training” to achieve its **closed-form** solution. **By setting different orders (i.e., D ’s), you can observe the change of the model.**

2. Given the problem in (1), implement the function “training_sgd” to learn the model via **stochastic gradient descent (SGD)**.
3. Given the above data matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_D]$, implement different normalization strategies:

$$\tilde{\mathbf{X}} = \mathbf{X} \text{diag}(\mathbf{a}), \quad \mathbf{a} = \left[\frac{1}{\|\mathbf{x}_1\|_p}, \dots, \frac{1}{\|\mathbf{x}_D\|_p} \right]^T, \quad p = 1, 2, \infty, \quad (2)$$

where $\mathbf{a}$ is called scaling vector, and $\text{diag}(\cdot)$ means converting a vector to a diagonal matrix.

4. Achieve a SGD algorithm based on the normalized data and implement a testing method for the model trained on the normalized data.
5. Given the model parameters derived by training on the normalized data, derive the model parameters corresponding to the original unnormalized data directly. (Are they the same with those obtained in Task 1?)

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2 Tasks

2 任务

1. 给定问题

$$\min_w \|y - Xw\|_2^2,$$

其中 $X = \{x_n^{d-1}\}_{n,d=1}^{N,D} \in \mathbb{R}^{N \times D}$ 是范德蒙德矩阵，实现函数 `training` 以求解其闭式解。通过设置不同的阶数（即不同的 D 值），你可以观察模型的变化。

2. 针对式 (1) 中的问题，实现函数 `training_sgd`，使用随机梯度下降（SGD）来学习模型。

3. 对于上述数据矩阵 $X = [x_1, \dots, x_D]$ ，实现不同的归一化策略：

$$\tilde{X} = X \operatorname{diag}(a), \quad a = \left[\frac{1}{\|x_1\|_p}, \dots, \frac{1}{\|x_D\|_p} \right]^T, \quad p = 1, 2, \infty,$$

其中 a 被称为缩放向量， $\operatorname{diag}(\cdot)$ 表示将一个向量转换为对角矩阵。

4. 基于归一化后的数据实现一个 SGD 算法，并实现一个测试方法，用于评估在归一化数据上训练得到的模型。

5. 给定在归一化数据上训练得到的模型参数，直接推导其对应于原始未归一化数据的模型参数。（它们是否与任务 1 中得到的参数相同？）
with those obtained in Task 1?)